UNIVERSITY OF CAPE TOWN

Department of Statistical Sciences

Experiment 1 [18 marks]

A sports scientist investigates whether warm-up routine affects sprint performance. She randomly assigns 24 athletes to one of three warm-up routines: Dynamic Stretching (DS), Static Stretching (SS), and No Warm-up (NW). Each athlete completes a 100m sprint and their time (in seconds) is recorded.

1.1 What is the treatment factor in this experiment? [2 mark]

Type: MCQ

(a) Warm-up routine ✓

(b) Sprint time in seconds

(c) Dynamic Stretching (DS), Static Stretching (SS), and No Warm-up (NW).

(d) Athlete

(e) The 100m sprint

1.2 How many treatments are there? Enter an integer number. [1 mark]

Type: Arithmetic / Short Answer

Answer: 3

1.3 How many replicates per treatment? Enter an integer number. [1 mark]

Type: Arithmetic / Short Answer

Answer: 8

1.4 What type of experimental design is this? [2 mark]

Type: Multiple Choice

(a) Factorial Randomised Complete Block Design

(b) Single Factor Completely Randomised Design ✓

(c) Factorial Completely Randomised Design

(d) Single Factor Randomised Complete Block Design

(e) Factorial Randomised Block Design

1.5 Why is it important that the sport scientist randomly assigns athletes to one of the three warm up routines? Limit your answer to a single sentence. [1 mark]

Type: Written Response

- *Random assignment ensures that any systematic differences between athletes (e.g. fitness level, experience) are spread out across treatment groups.*
- *Any answer similar in spirit — e.g. preventing or accounting for confounding variables, or ensuring a fair/unbiased comparison of the warm-up routines.*
- *Removes bias on behalf of the researcher/sport scientist.*

1.6 In the ANOVA model: $Y_{ij} = \mu + A_i + e_{ij}$, the A_i term represents the variation due to individual differences between subjects within the same group. [1 mark]

Type: True or False

Answer: False. The treatment effects represent variation between treatment means, due to treatment effect. The error term captured individual variation.

1.7 The following partial ANOVA table was obtained. Complete the missing values (a)–(e). [5 marks, 1 mark each]

Type: Multiple short answers

Source	SS	df	MS	F	p-value
Treatment	(a)	2	60	(d)	< 0.01
Residuals	(b)	(c)	10		
Total	330	(e)			

Answers:

(a) 120 [SS_{Treat} = MS×df = 60×2]

(b) 210 [330–120]

(c) 21 [N–a = 24–3]

(d) 6 [60/10]

(e) 23 [N–1 = 24–1]

1.8 The treatment factor accounted for the largest portion of variation in the data. [1 mark]

Type: True or False

Answer: False. Decomposition of sums of squares indicated the residuals accounted for the largest proportion of variation.

1.9 For the following question, make sure to answer each of the three parts:

- Based on the ANOVA table, state the null and alternative hypotheses (in words, and without referring to ANY symbols, i.e., not typing 'mu' to represent a mean, etc.).
- Decide whether to reject or not reject the null hypothesis by supplying evidence for your decision, and
- Interpret the decision in context of the experiment

Type: Written Response

Mark with error: If a student gets the decision wrong but their conclusion is consistent with their (incorrect) decision, award the mark for the conclusion.

Model answer:

- ***H_0*:** *All treatment means are equal (or equivalently, all treatment effects = 0) / warm-up routine does not affect sprint performance.* ***[0.5]***
- ***H_1*:** *At least one treatment mean differs / at least one treatment effect is not equal to zero / warm-up routine affects sprint performance.* ***[0.5]***
- *Since $p < 0.001 < 0.05$* ***[0.5]****, we reject H_0* ***[0.5]****.*
- *There is significant evidence that warm-up routine affects sprint time.* ***[1]***

*Marker's note**:** No marks for clearly wrong hypotheses. Saying "none of the warm-up routines affect performance" is wrong; the hypothesis is that they have the same means; they can all have the same mean sprint performance. The effect is that there is a difference in means. Same for the alternative: saying "at least one warm-up routine affects sprint performance" is wrong.*

1.10 Specify the sampling distribution of the test statistic by giving its name and appropriate degrees of freedom. [1 mark]

Type: Written response

Answer: F distribution with 2 and 21 degrees of freedom. [0.5 for F, 0.5 for correct degrees of freedom]

Experiment 2 [11 marks]

A coffee company wants to know whether roasting profile (Light, Medium, Dark) affects the caffeine concentration (mg/100ml) of brewed coffee. The company sources its beans from five different origins: Ethiopian, Colombian, Brazilian, Kenyan, and Guatemalan. For each origin, three separate samples of beans are prepared , one for each roasting profile, and the caffeine concentration of the resulting brew is measured. After conducting the experiment and storing the data in *coffee.csv*, the analyst used the following model:

$CaffeineConcentration_{ij} = \mu + RoastingProfile_i + BeanOrigin_j + e_{ij}$

Load the data in R and answer the following questions.

2.1 What is the experimental unit in this experiment? [1 mark]

Type: Short Answer

Answer: A batch/sample of beans / brewed coffee / coffees to be made / coffee cups

2.2 Why is bean origin included in the experiment? . [1 mark]

Type: Written Response

Answer: To account for or remove variability between bean origins that might influence caffeine levels. Any answer similar in spirit: e.g. to reduce experimental error variance, or to control for a known variable that affects the response but is not of interest.

2.3 Which of the following are assumptions of the ANOVA model used in this experiment? Select ALL that apply. [2 marks]

Type: Multi-Select

(a) Errors are independently distributed ✓

(b) Errors are normally distributed ✓

(c) Equal population variances across treatment groups ✓

(d) Treatment effects sum to zero.

(e) No interaction between treatment and block ✓

2.4 What is the overall mean caffeine concentration? [1 mark]

Type: Short Answer

Answer: 9.40/9.4

2.5 Give the definition of a treatment effect in context of the experiment (in words, not symbols). [1 mark]

Type: Written response

Answer: *The treatment effect is the difference between the mean caffeine concentration of a treatment and the overall caffeine concentration. Half a mark for stating this in general terms only (i.e. "treatment mean minus overall mean" without reference to the response variable, caffeine concentration).*

2.6 What is the fitted value (predicted caffeine concentration) for the Dark roast with Colombian beans? [1 marks]

Type: Short Answer

Answer: $\hat{\mu} + \hat{A}_{Dark} + \hat{B}_{Colombian} = 9.40 + 2.32 + 0.67 = 12.39$

2.7 True or False: It was not necessary to include bean origin in the model. A completely randomised design would have been sufficient. [1 mark]

Type: True or False

Answer: True. The F-value for block effects is close to 1, suggesting that the blocking variable did not reduce the experimental error variance.

2.8 What is the MSE from the ANOVA table? [1 mark]

Type: Short Answer

Answer: MSE = 0.86.

2.9 What does the MSE estimate? [1 mark]

Type: Written Response

Answer:

Full mark:

- *It estimates the experimental error variance* σ^2.
- *Within-group variability.*
- *How much variation is not explained by the treatment means / differences in treatment means.*
- *Variance of the residuals.*

Half mark:

- *How much variation is not explained by the mean (without specifying treatment means).*

2.10 In the formula of the standard error for the treatment means, what is the value of “n”? Enter an integer number. [1 mark]

INSERT FORMULA: \sqrt{\frac{\sigma^2}{n}}

Type: Short answer
Answer: 5

Experiment 3 [11 marks]

A researcher investigates whether study method (Flashcards, Summarising, Practice Tests) and background noise (Quiet, Music) affect exam scores. This is a 3×2 factorial experiment in a CRD. There are 8 replicates per treatment combination (N = 48).

Load the dataset *exam_study_data.csv* into R. Convert variables to factors if necessary. Fit the appropriate model in R. Use *model.tables()* and *summary()* to answer the questions below.

3.1 Why is replication (beyond allowing quantification of experimental error variance) important for factorial experiments? [1 mark]

Type: Written response

Answer: Replication in factorial experiments ensures that the interaction effect can be estimated / separates the error term from the interaction effect.

3.2 What is the ratio of largest to smallest standard deviation across the six treatment groups?

Type: Short answer

Answer: 2.35

3.3 Based on the ratio of largest to smallest standard deviation, the assumption of equal population variance is reasonable.

Type: True or false

Answer: True (less than 5)

3.4 What is the estimated interaction effect for the Practice Tests + Music treatment? [1 mark]

Type: Short Answer / Written Response

Answer: 4.82 [1 mark]

3.5 At α = 0.05, which effects are statistically significant? Select ALL that apply. [2 marks]

Type: Multi-Select

(a) Study Method ✓

(b) Noise

(c) Study Method × Noise interaction ✓

(d) Residuals

(e) All effects are statistically significant.

3.6 Construct and investigate the interaction plot. Interpret the interaction between Study Method and Noise in context. What does the interaction pattern suggest? [2 marks]

Type: Written Response

Answer: The effect of background noise on exam scores depends on the study method used.

3.7 True or False: In this experiment, we can interpret the main effects of Noise in isolation even though there is an interaction present. [1 mark]

Type: True or False

Answer: False. When an interaction is present, interpreting main effects alone can be misleading because the effect of one factor depends on the level of the other.

3.8 Construct a plot of the residuals versus the fitted values. Name one assumption that can be checked by inspecting this plot and decide whether it has been met. [2 marks]

Type: Written Response

Answer:

- *Whether the variance is constant (homoscedasticity); the assumption appears to be met.*
- *Whether the errors have a mean of zero; this is reasonable based on the plot.*
- *Normality of errors; acceptable as well.*

Marking: 1 mark for correctly identifying the assumption, 1 mark for stating it has been met (both required for full marks).